

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Course Name	Computer Networks	Course Code	CSA07
Course Outcome	CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)		
Assessment Tool	Concept Mapping (Annexure A)	Weightage	30%
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Question 1

A partially completed concept map is given below: Application → Presentation → _____ → Transport → Network → _____ → Physical. Analyze the missing layers and explain how their functions are essential for reliable communication.

Application → Presentation → Session → Transport → Network → Data Link → Physical

Answer:

The two missing layers in the OSI concept map are the **Session Layer** (between Presentation and Transport) and the **Data Link Layer** (between Network and Physical).

Session Layer: This layer is responsible for establishing, managing, and terminating communication sessions between two applications. It handles dialog control (full-duplex or half-duplex communication), synchronization by inserting checkpoints into the data stream, and session recovery in case of a failure. Without the session layer, applications would not be able to keep track of ongoing conversations, and any interruption would force communication to restart from the beginning, reducing reliability.

Data Link Layer: This layer is responsible for node-to-node delivery of data frames within the same local network. It performs framing, physical addressing (MAC addressing), error detection using techniques such as CRC, and flow control between adjacent nodes. It converts the raw bit stream received from the Physical layer into recognizable frames and passes them to the Network layer.

Importance for reliable communication:

- The Session layer ensures that data exchange between two communicating applications remains synchronized and organized, allowing recovery without total data loss if a session is interrupted.
- The Data Link layer ensures that frames are correctly delivered across a single physical link by detecting transmission errors and controlling the rate of data flow, preventing frame loss or corruption.
- Together, all seven layers work in a hierarchical, complementary manner — each layer adds its own header/trailer during encapsulation and removes it during decapsulation — so that data delivered by the Physical layer is reliably transformed step-by-step into meaningful information for the Application layer.
- This layered concept map demonstrates that reliable communication is not the responsibility of a single layer; instead, it emerges from the cooperative interaction of session management, transport reliability, routing, and error-free local delivery.

Question 2

Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Transport Layer → Segment | Network Layer → Packet | Data Link Layer → Frame | Physical Layer → Bits

Answer:

The concept map links each OSI layer to the specific Protocol Data Unit (PDU) it creates and processes during the encapsulation process:

OSI Layer	Data Unit (PDU)	Function
Transport Layer	Segment	Divides application data into segments; adds port numbers and sequence numbers
Network Layer	Packet	Encapsulates the segment with logical (IP) addressing to enable routing across networks
Data Link Layer	Frame	Encapsulates the packet with MAC addresses and error-checking information for local delivery
Physical Layer	Bits	Converts the frame into a raw stream of bits (electrical, optical, or radio signals)

Analysis of transformation supporting transmission:

- This step-by-step transformation is called **encapsulation**. As data travels down the OSI stack from sender to the physical medium, each layer wraps the data unit received from the layer above with its own header (and sometimes trailer), converting it into a new, layer-specific data unit.
- At the receiving end, the reverse process, **decapsulation**, occurs: bits are reassembled into frames, frames into packets, packets into segments, and segments into the original application data.
- This transformation supports transmission because each data unit carries only the control information relevant to its own layer — for example, a Frame carries MAC addresses for local delivery, while a Packet carries IP addresses for global routing — preventing unnecessary overhead and ensuring each layer performs its task independently.
- If any transformation fails (e.g., a frame is corrupted or a segment's checksum does not match), the concerned layer can detect the fault and either discard, retransmit, or request correction, thereby supporting reliable end-to-end transmission.

Question 3

Given the concept map: Data Link → MAC Address → Local Delivery; Network → IP Address → Global Delivery. Explain how these relationships ensure complete data delivery across networks.

Data Link Layer → MAC Address → Local Delivery | Network Layer → IP Address → Global Delivery

Answer:

This concept map illustrates the two complementary addressing mechanisms that together enable complete, end-to-end delivery of data across interconnected networks.

Data Link Layer – MAC Address – Local Delivery: Every network interface card has a unique, physically burned-in MAC (Media Access Control) address. The Data Link layer uses this address to deliver frames between devices that exist on the same local network segment (LAN). MAC addressing is only meaningful within the local network and changes at every hop as a frame crosses from one network segment to another (through routers).

Network Layer – IP Address – Global Delivery: The Network layer uses logical IP addresses, which are hierarchical and can be structured by network administrators. IP addressing allows a packet to be routed across multiple interconnected networks (the Internet) from the source host to the destination host, regardless of the underlying physical network technology.

How the two relationships together ensure complete delivery:

- The **IP address remains constant** throughout the packet's journey from source to destination, providing global, end-to-end logical addressing.
- The **MAC address changes at every hop** — each router examines the destination IP address, determines the next hop, and re-encapsulates the packet in a new frame with the MAC address of the next device in the path.
- This layered address translation (using protocols such as ARP to map an IP address to a MAC address) means that global delivery (Network layer) is achieved through a sequence of successful local deliveries (Data Link layer) at each intermediate network segment.
- Hence, the concept map shows that reliable, complete communication depends on the cooperative relationship between logical addressing for routing decisions and physical addressing for the actual hop-by-hop transfer of frames.

Question 4

A concept map shows: Error Control → Data Link Layer → Error Detection; Error Control → Transport Layer → Error Correction. Analyze how the interaction between these layers improves reliability in data transmission.

Error Control → Data Link Layer → Error Detection (CRC, checksum, parity) | Error Control → Transport Layer → Error Correction (ACK/NACK, retransmission)

Answer:

Error control is distributed across multiple OSI layers, with each layer contributing a different aspect of reliability rather than duplicating the same function.

Data Link Layer – Error Detection: This layer focuses on detecting errors introduced during transmission over a single physical link, typically caused by noise, interference, or signal attenuation.

Techniques such as Cyclic Redundancy Check (CRC), checksums, and parity bits are appended to each frame. If the receiving node recalculates the check value and finds a mismatch, the frame is identified as corrupted and is usually discarded at that hop.

Transport Layer – Error Correction: This layer is responsible for end-to-end reliability across the entire path from source to destination, which may span multiple links and intermediate networks. Using mechanisms such as sequence numbers, acknowledgements (ACK), negative acknowledgements (NACK), timers, and retransmission (as implemented in TCP), the Transport layer can detect missing or damaged segments and request retransmission until correct, complete data is received.

Interaction between the two layers:

- The Data Link layer acts as a first line of defense, catching and discarding corrupted frames close to the point where the error occurred, which prevents faulty data from propagating further into the network.
- However, since the Data Link layer typically only detects and discards errors (it does not guarantee recovery across multiple hops), the Transport layer provides the second line of defense by ensuring that if a segment never arrives (because a frame was dropped somewhere along the path), the sender is eventually notified and retransmits the missing data.
- This two-tier interaction — local error detection at the Data Link layer combined with end-to-end error correction at the Transport layer — significantly improves overall reliability, because errors are handled efficiently at the layer best suited to fix them, minimizing unnecessary retransmissions over long distances while still guaranteeing that the destination ultimately receives an accurate, complete copy of the data.

Question 5

Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Answer:

Concept Map – Web Browsing across OSI Layers:

OSI Layer	Role in Web Browsing (e.g., loading a website)
Application	HTTP/HTTPS request generated by the browser (e.g., GET request for a webpage).
Presentation	Data formatting, encryption/decryption (TLS/SSL for HTTPS), compression of the web content.
Session	Establishes and maintains the session between the browser and the web server for the duration of the page load.
Transport	TCP establishes a reliable connection (three-way handshake), segments the HTTP request, ensures ordered delivery.
Network	IP addressing and routing of packets from the client to the web server across the Internet.
Data Link	Framing and MAC addressing for delivery across the local network (e.g., Wi-Fi router or Ethernet switch).
Physical	Actual transmission of bits as electrical, optical, or radio signals over the physical medium.

Answer:

Web browsing is a practical example that demonstrates how all seven OSI layers cooperate to deliver a single user action — loading a webpage.

Analysis of layer failure and its cascading effect:

- **Application layer failure** (e.g., malformed HTTP request or unreachable server): the browser cannot generate or interpret the request, so no webpage content can be retrieved, even if all lower layers are functioning correctly.
- **Presentation layer failure** (e.g., TLS/SSL handshake failure): encrypted HTTPS content cannot be decrypted, causing the browser to display a security warning and block the page, even though the connection itself may be intact.
- **Session layer failure**: the ongoing browsing session may be lost, forcing the user to log in again or restart the page load from scratch.
- **Transport layer failure** (e.g., TCP connection cannot be established or segments are repeatedly lost): the browser will show a 'connection timed out' error, since reliable delivery of the HTTP data cannot be guaranteed.
- **Network layer failure** (e.g., incorrect routing or unreachable IP address): packets cannot find a path to the destination server, resulting in a 'destination unreachable' error.
- **Data Link layer failure** (e.g., Wi-Fi disconnection or switch fault): frames cannot be delivered even within the local network, so no data can leave the client device at all.
- **Physical layer failure** (e.g., damaged cable, no signal): there is a complete loss of connectivity, and none of the upper layers can function since no bits can be transmitted.
- This concept map shows that OSI layers are strictly interdependent: a failure at any single layer breaks the entire communication chain because each layer relies on the correct functioning of the layer below it to deliver its own service to the layer above. This reinforces why network troubleshooting is typically performed layer-by-layer, starting from the Physical layer upward.

Question 6

A network issue is represented in the concept map: Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗. Analyze the problem and explain how the concept map helps in troubleshooting.

Physical (OK) → Data Link (OK) → Network (FAIL) → Transport (FAIL) → Application (FAIL)

Answer:

Analysis of the problem: The concept map indicates that the Physical layer and Data Link layer are functioning correctly — meaning the physical cabling/wireless connection is active and the local device can successfully send and receive frames within its local network segment (e.g., the device is connected to the switch/router and has link connectivity). However, the failure begins at the Network layer and cascades upward to the Transport and Application layers.

Likely causes at the Network layer:

- Incorrect or missing IP address configuration (e.g., no valid IP obtained from DHCP).
- Incorrect default gateway or subnet mask configuration, preventing packets from being routed outside the local network.
- Routing failure or an unreachable next-hop router.
- Firewall or ACL rules blocking IP-level traffic.

Why Transport and Application layers also fail: Since the Transport layer (TCP/UDP) depends on the Network layer to deliver packets to the correct destination IP address, if the Network layer cannot route or address packets correctly, no segments can reach the destination, so TCP connections cannot be established. Consequently, the Application layer (e.g., a browser or email client) cannot receive any usable data and reports connectivity errors, even though the underlying physical and local network connection is completely healthy.

How the concept map helps in troubleshooting:

- The map visually pinpoints exactly where the chain of connectivity breaks — between Data Link and Network — allowing a network engineer to immediately focus diagnostic effort on IP addressing, routing tables, and gateway configuration instead of wasting time checking cables or switch ports.
- This reflects the standard **bottom-up troubleshooting approach** used in real-world networking: verify Physical connectivity first (cable/link lights), then Data Link (switch port status, MAC table), then Network (ping gateway, check IP configuration, traceroute), and so on.
- Because the concept map clearly shows which layers are functioning (✓) and which are not (✗), it eliminates guesswork, reduces troubleshooting time, and allows the engineer to isolate the root cause to a specific layer rather than treating the failure as a vague, whole-network problem.
- This demonstrates the practical value of concept mapping in network diagnostics: by representing the OSI model as an interconnected map with layer-wise status indicators, faults can be logically isolated and resolved systematically.